

2 Static Analysis

All elements of a mechanism, links and joints, must be carefully designed so that the mechanism runs smoothly, efficiently and reliably while withstanding the forces encountered in operation. We have an intuitive sense of the terms “smoothly”, “reliably” and “efficiently” but to optimize and evaluate all the components of the mechanism and its actuator, we need to quantify what these terms mean. In our definition, a smoothly running mechanism is one that does not bind. To begin with, we know that a mechanism can bind if, as it goes through the motion cycle, the angles between various links change too suddenly or too much. That is, at certain points, these angles may become too large or too small. We can predict this by performing a position analysis to model various angles during the motion cycle.

In this chapter, we’ll explore how the joints and elements are carefully selected and sized so that the mechanism does not bind and fail in operation. In order to select the proper types of bearings (configuration and size), we must first determine the types of forces that are encountered by the joints. Likewise, we need to look at the forces transmitted through the joints that are imparted on the link members. We must then ensure that the stresses within each link fall below the failure stress threshold through choice of proper materials and cross-sectional sizes. Therefore, we must first understand all the forces and motions transmitted by the mechanism. To do this, different types of analysis are performed – static, kinetostatic, kinematic and dynamic, depending on what is initially known about the mechanism.

In static analysis, we are primarily concerned with the mechanical advantage of the mechanism. That is, what input force is required to balance the external load at each position of the mechanism in motion. In position analysis we focus on the geometric and kinematic inversions of a mechanism. In kinetostatic analysis the motion of the mechanism is given and we seek the required input force to sustain the assumed motion. In kinetostatic analysis, the dimensions, mass, position, velocity and acceleration of all the links and the external loading are known. We must seek the resulting joint forces and the input torque that is required to drive the mechanism so that it meets the prescribed velocity and acceleration characteristics. Thus, in kinematic and kinetostatic analysis, we treat the position of the input (actuator) link as the independent variable and seek the other quantities. As for dynamic analysis, the independent variable is time. We seek to determine the actual motion by taking into account inertia forces and external loading and solving differential equations of motion.

Static Analysis:

analysis of a mechanism to determine the mechanical advantage and the geometric and kinematic inversions

Kinetostatic

Analysis: analysis of a mechanism to determine the required input force for a given motion

Dynamic Analysis:

analysis of a mechanism to determine the motion over time, given the inertia and external forces on the mechanism

2.1 POSITION ANALYSIS

To begin with, we will focus on the position and static analyses. Much of the study of position analysis concentrates on four-bar mechanisms, but the procedures are also applicable to more complex mechanisms. In Chapter 1 we focused primarily on degrees of freedom. We found that the four-bar linkage has a single degree of freedom. There are other characteristics of four-bar linkages that distinguish them from one another which we should also know so that we can choose an appropriate type for a given application. These include Grashof criteria, the concept of inversion, dead center position circuits, and transmission angles.

Grashof Criteria

Grashof Criteria:
the sum of the shortest and longest links of a planar four-bar linkage cannot be greater than the sum of the remaining two links if there is to be a continuous relative rotation between the pairs of links

A four-bar linkage may take the form of a so-called crank-rocker, double-rocker or double-crank (drag-link) linkage, depending on the *range of motion* of the two links in relation to the ground link. In Figs. 2.1 to 2.4 four different possibilities are illustrated. The input crank of the *crank-rocker* type (Fig. 2.1) can rotate continuously through 360° while the output link just “rocks” (or oscillates). Both the input and output links of the *double-crank* or *drag link mechanism* (Fig. 2.2) make complete revolutions, while the double-rocker has limited rotations of both the input and output links (Fig. 2.3) In the parallelogram linkage (Fig. 2.4), where the length of the input link equals that of the output link and the lengths of the coupler and ground link are also equal, both the input and output link may rotate entirely around or switch into a crossed configuration called an *anti-parallelogram* linkage.

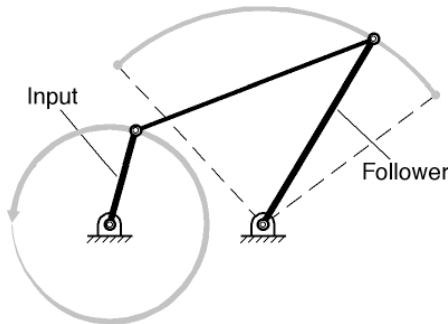


Figure 2.1 Crank and rocker

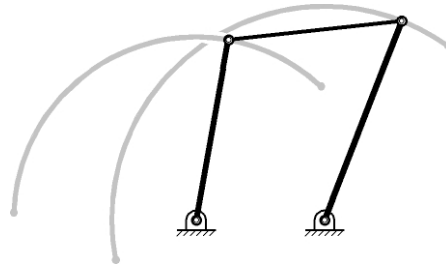


Figure 2.2 Drag-link or double-crank

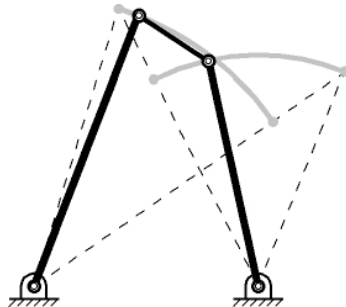


Figure 2.3 Double-rocker

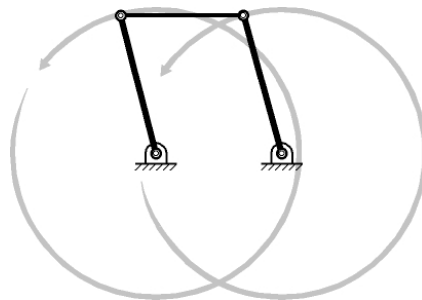


Figure 2.4 Parallelogram linkage

As one might guess, any particular four-bar linkage can be recognized as one of these types, depending on some relationship of its link lengths. Grashof criteria describe this relationship. The Grashof's criterion states that: the sum of the shortest and longest links of a planar four-bar linkage cannot be greater than the sum of the remaining two links if there is to be a continuous relative rotation between the two links. If we identify the longest link as l , the shortest as s , and the remaining two as p and q , the following relationships [86] are valid (Fig. 2.5):

Change Point Condition: Where the centerlines of all links in a mechanism can become collinear.

1. If $l + s < p + q$, four possibilities or Grashof mechanisms exist.
 - a. A crank-rocker mechanism is obtained when the shortest link is the crank, the frame being either adjacent link (Fig. 2.7a).
 - b. A double-crank (drag-link) results when the shortest link is the frame (Fig. 2.7b).
 - c. A rocker-crank mechanism is formed when the shortest link is the follower (Fig. 2.7c).
 - d. A double-rocker mechanism results when the link opposite the shortest is the frame (Fig. 2.7d).
2. If $l + s > p + q$, four non-Grashof triple-rocker mechanisms result depending on which link is the frame (Fig. 2.7e). Continuous relative motion is not possible for this case.
3. If $l + s = p + q$, the four possible mechanisms of case 1 exist, but they all suffer from the *change-point condition*: The centerlines of all links can become collinear, creating momentarily a second degree of freedom. Also a toggle condition (which occurs when the input and coupler are lined up) will be present.



Figure 2.5 Toggling Condition

The parallelogram linkage and the deltoid linkage are special cases of *change point mechanisms* where $l = q$, and $s = p$. In the parallelogram linkage the short links are separated by a long link (Fig. 2.4). All parallelograms are double-cranks, but must be controlled through the change points. This is the only four-bar capable of producing parallel motion of the coupler, and all paths traced by the coupler are circular arcs.

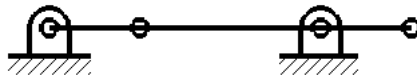


Figure 2.5a Parallelogram linkage in change-point condition

The *deltoid* has two adjacent equal-length short links connected to two adjacent equal-length longer links. With a long side as the frame, a crank-rocker is possible; a short side as the frame may give a double-crank in which the short rotating link makes two turns to the longer link's one (a *Galloway mechanism*). Again, this linkage is vulnerable to a change-point condition.

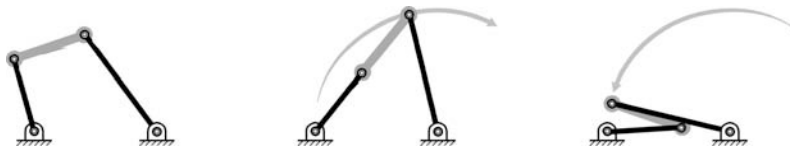


Figure 2.5b Deltoid linkage approaching change-point condition

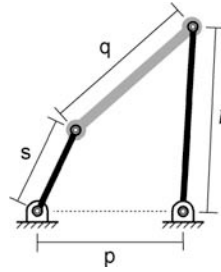


Figure 2.6 Four-bar linkage: shortest link s , longest link l , intermediate length links p and q .

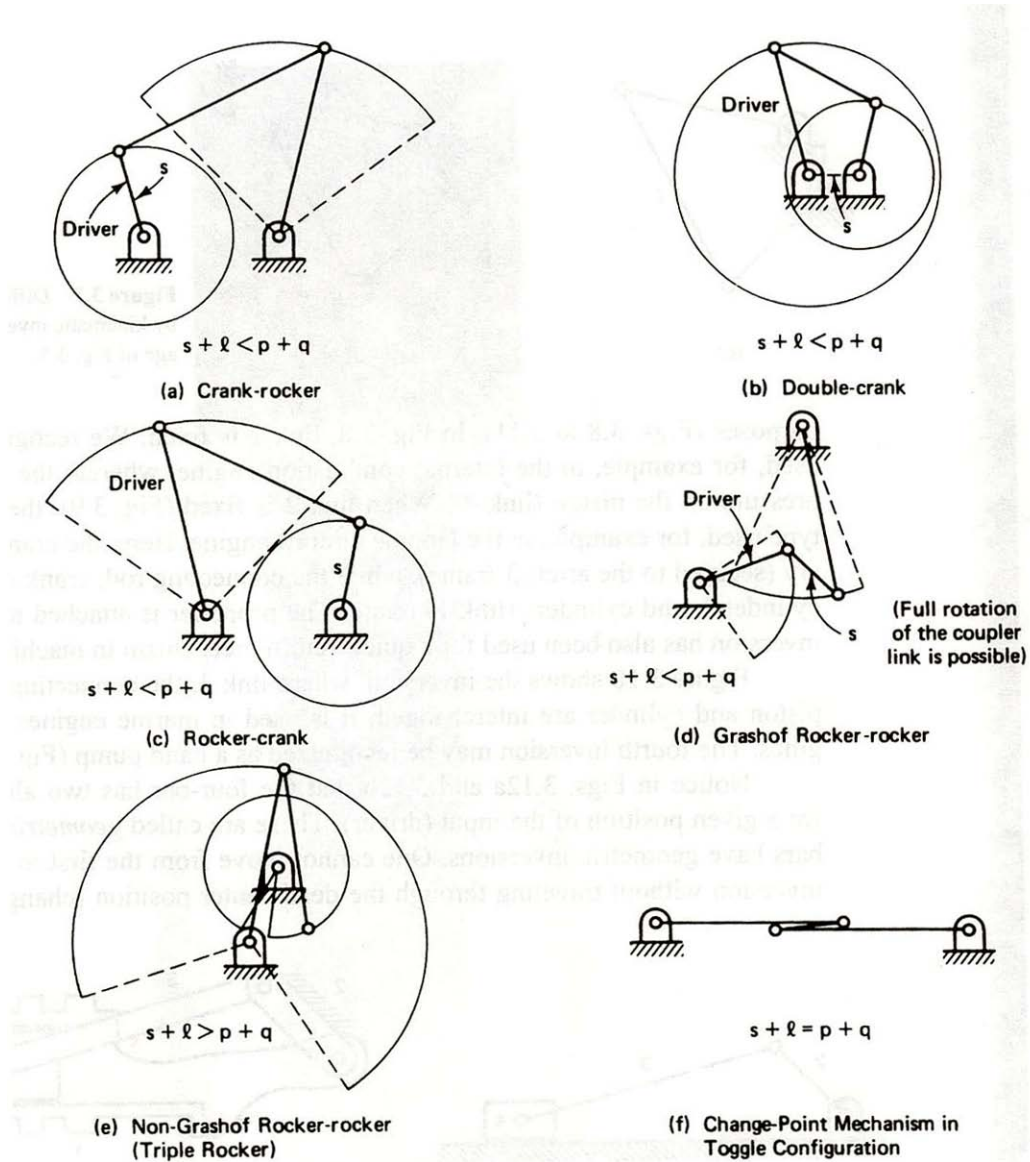


Figure 2.7 Grashof and non-Grashof types of four-bar mechanisms [24].

Kinematic Inversions

Kinematic Inversion: the process of fixing different links in a kinematic chain to create different mechanisms.

Figures 2.8a – 2.8d are the Grashof four bars of Case 1, where $l + s < p + q$. The same four link Grashof configuration chain can be either of the cases under 1, depending on which link is specified as the frame (or ground). *Kinematic inversion* is the process of fixing different links of a chain to create different types of mechanisms. Note that the relative motion between links of a mechanism does not change in different inversions. This property will be used to good advantage throughout in this book. All other linkages can also have kinematic inversions. For example, inversions of the slider-crank mechanisms are used for different purposes (Figs. 2.9 to 2.12). In Fig. 2.9, link 1 is fixed. We recognize this linkage from the internal combustion engine, wherein the input force is the gas pressure on the piston (link 4). When link 2 is fixed (Fig. 2.10), the linkage becomes the type used, for example, in the Gnome aircraft engine. Here, the crankshaft is held stationary (secured to the aircraft frame), while the connecting rod, crank case (integral with the cylinders), and cylinders (link 1) rotate. The propeller is attached to the crank case. This inversion has also been used for a quick-return mechanism in machine tools.

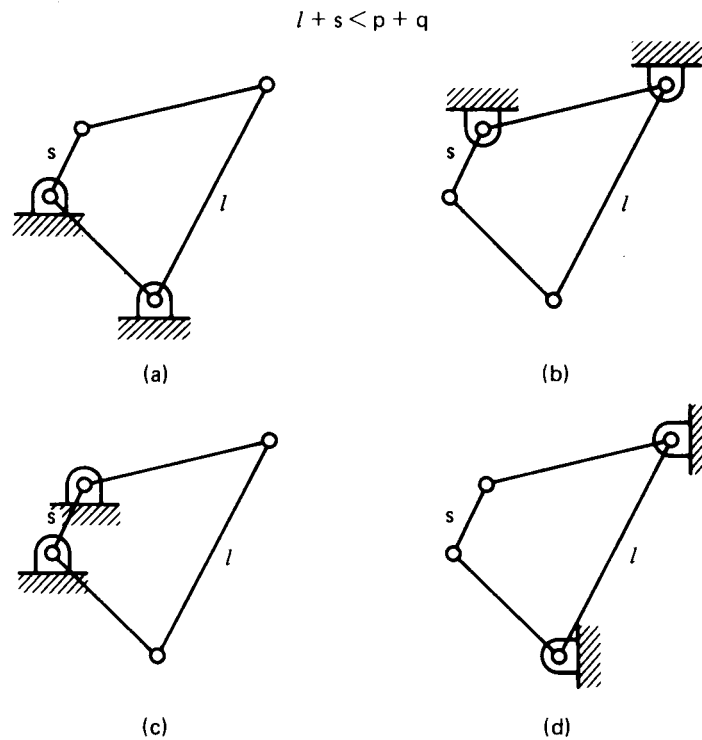


Figure 2.8 Different mechanisms formed by kinematic inversions of the four-bar linkage of Fig. 2.6

Figure 2.11 shows the inversion, where link 3, the connecting rod, is fixed, and the piston and cylinder are interchanged; it is used in marine engines and in toy steam engines. The fourth inversion may be recognized as a hand pump (Fig. 2.12).

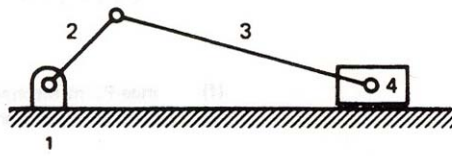


Figure 2.9 Slider-crank

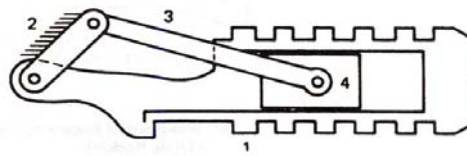


Figure 2.10 Kinematic inversion of the slider-crank: rotating slide.

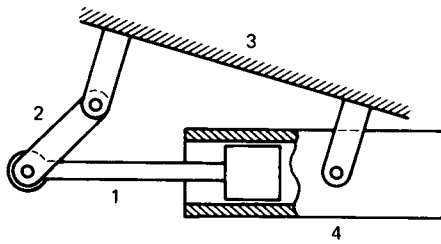


Figure 2.11 Kinematic inversion of the slider-crank: oscillating slide.

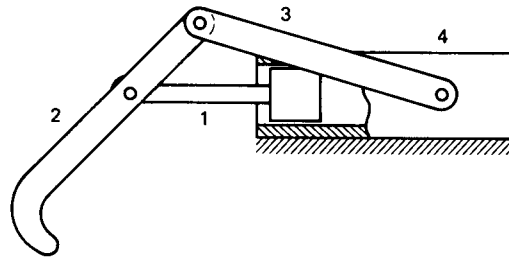


Figure 2.12 Kinematic inversion of the slider-crank: stationary slide.

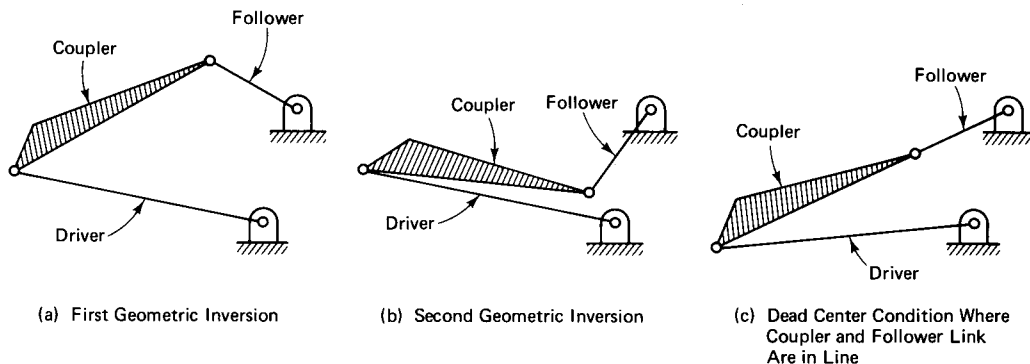


Figure 2.13 Two geometric inversions of a four-bar linkage.

Geometric Inversion

Geometric Inversion:
alternative configuration of a linkage

Notice in Figs. 2.13a and 2.13b that the four-bar has two alternate configurations for a given position of the input (driver). These are called *geometric inversions*. All four-bars have geometric inversions. One cannot move from the first to the second geometric inversion without disassembling or traveling through the *dead center position* (Fig. 2.13c). But both ranges of motion in Figs. 2.13a and 2.13b can be reached without taking the four-bar apart if the dead center can be negotiated. One can make use of a dead center position as is done with a rear seat linkage (Fig. 2.14). This is a sketch of a four-bar linkage that guides the motion of both the seat back and the cover plate pan of a 1986 Ford Mustang. To lower the seatback, one grasps the seatback (input link) and rotates it forward (counterclockwise in this view). About the time the seatback is parallel to

the lower portion of the seat, the coupler and output link (cover plate) are in line. One then pushes the cover plate down (counterclockwise) through the dead center into a stable latched position.

In the case of all Grashof four-bars, there are two phases of possible motion that can only be obtained by physically disconnecting the joint between the coupler and follower links. These are called *separate circuits*, as illustrated in Fig. 2.15. Non-Grashof mechanisms have only a single circuit (Fig. 2.16) containing both geometric inversions.

Crank-rocker and double-crank mechanisms never reach a dead center position; so the two geometric inversions always fall on the two separate circuits (Fig. 2.15). Conversely, each circuit is composed of the same geometric inversion. Rocker-cranks or double-rockers (Fig. 2.16) have two dead center positions on both circuits and each circuit has a distinct range of motion of the driver.

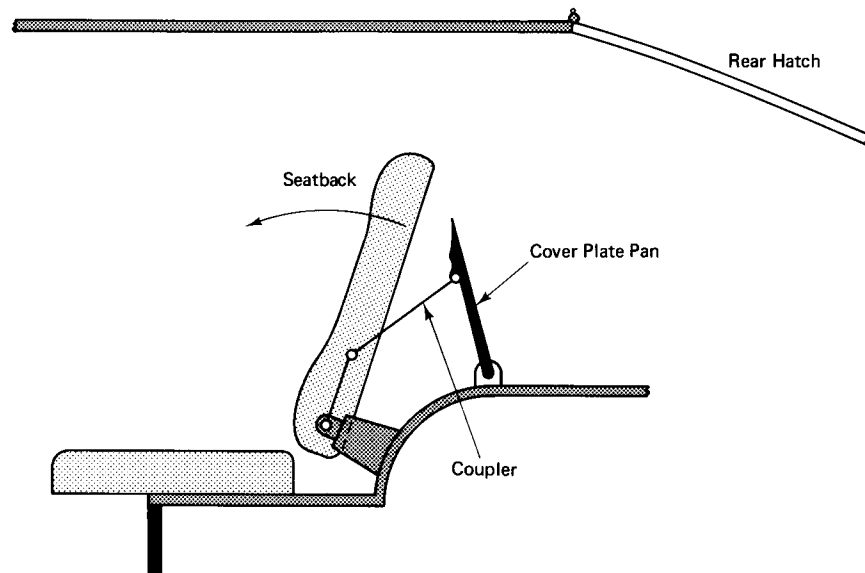


Figure 2.14 Sketch of a 1986 Ford Mustang rear seat linkage moves through the dead center position in the seat down position. (supplied by Dr. Tom Chase, University of Minnesota, along with Figs. 2.13, 2.15 and 2.16).

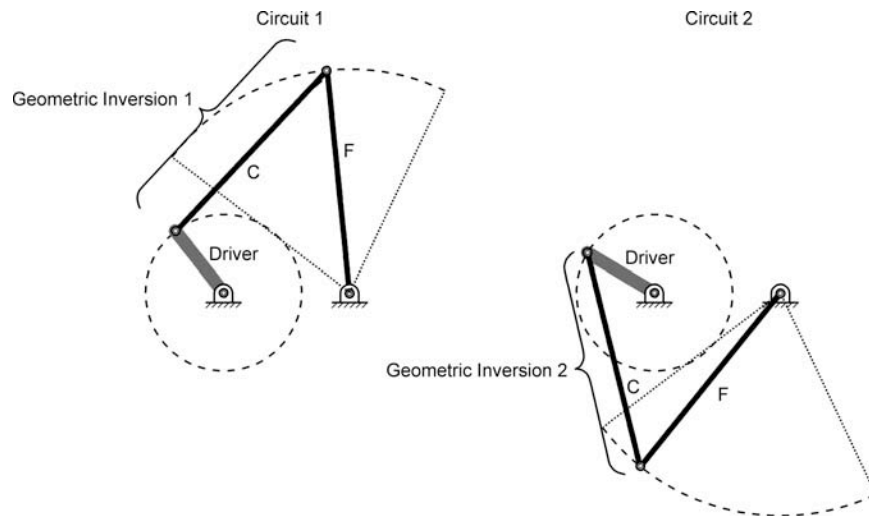


Figure 2.15 If the four-bar cannot reach a dead center (non-Grashof) position, each geometric inversion will fall on a distinct circuit.

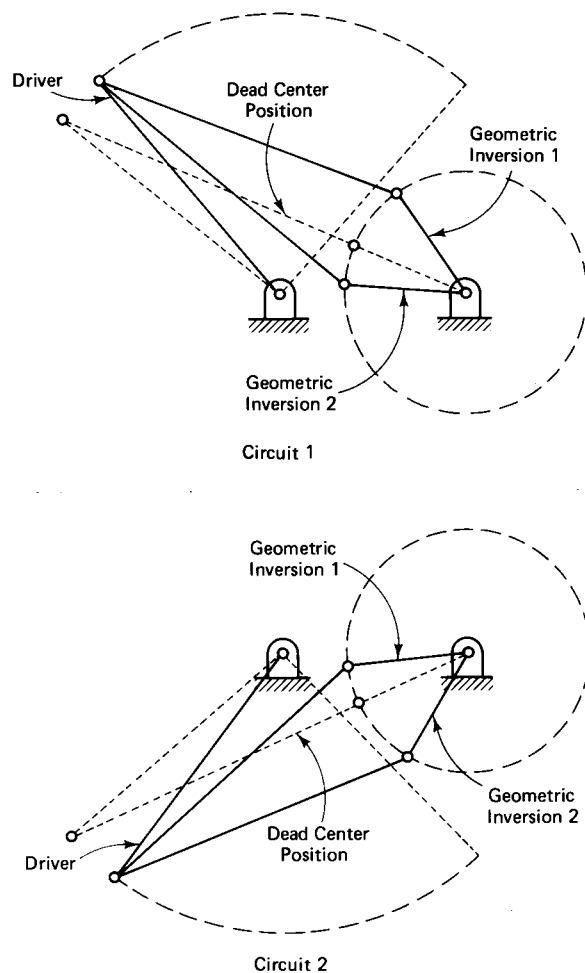
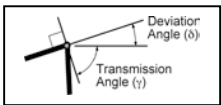


Figure 2.16 If the four-bar can reach a dead center (non-Grashof) position, both geometric inversions occur on

Besides knowing the extent of rotation of the input and output links, it is useful to have an understanding of how well a mechanism might run before actually building it. Hartenberg and Denavit [86] mention that “run is a term that more formally means the effectiveness with which motion is imparted to the output link. It implies smooth operation in which a maximum force component is available to convey a torque or force, whatever the case might be, to an output member.” The resulting output force or torque is not only a function of the geometry of the linkage. It is also the result of dynamic or inertia forces (see Chap. 5), which are often several times as large as the static forces and can act in quite different directions. For the analysis of low-speed operation or for an easily obtainable index of how any mechanism might run at moderate speeds, the concept of the transmission and deviation angle is extremely useful.

Deviation Angle is the angle between the direction of the **force vector** and the direction of the **velocity vector** of the output. Ideally, we want this angle to be zero so that all of the force is employed to impart the motion. If this angle is not zero, then the component of the force that is perpendicular to the motion is not only “not-contributing” to the motion but is causing friction and binding in the joint. As the linkage goes through the motion cycle, this angle can vary. In order for the mechanism to “run smoothly”, the deviation angle should not exceed say 30° as a rule of thumb. The direction of the force imparted on the output link is determined by the orientation of the link connected to the output link that is transmitting this force. In Figure 2.17, the floating link 3 is transmitting the force to the output link 4.



Transmission angle is defined as the smaller (acute) angle between the direction of the **velocity difference** vector of the floating link and the direction of the **absolute velocity** of the output link, both taken at the point of connection. Simply stated, transmission angle is the angle complementary to the deviation angle; that is $\text{transmission angle} = 90^\circ \text{ minus the deviation angle}$. The transmission angle γ is illustrated in the four-bar linkage of Figs. 2.16 and 2.17. The velocity-difference vector, denoted as $\mathbf{V}_{BA}$ (velocity of point B relative to point A), is perpendicular to the floating link (link 3 in this case), while the absolute velocity of the output is perpendicular to link 4.

The sum of the transmission angle and deviation angle is 90° if and only if the link is a two-force member.

As shown in Figure 2.16, the *deviation angle* δ is the smallest angle between the direction of the static force $\mathbf{F}_{34}$, transmitted through the floating link, and the absolute velocity of the output link, $\mathbf{V}_B$, at the point of connection. The direction of the static force of the floating link is along the line of its pin joints, since the link is a two-force member (due to the absence of any other force on the link and the assumption of frictionless pin joints at its ends). The *pressure angle* used in cam and cam-follower systems is equivalent to the deviation angle. The authors prefer to use the deviation angle δ rather than transmission angle γ , because it is quicker to find the absolute velocity and the static force.

Notice that $\gamma + \delta = 90^\circ$. This relation is true whenever the coupler link has just the two opposite joint forces acting on it (a *two-force member*). This relation does not hold true when there is a *three-force member*. The optimum transmission angle is 90° while the optimum deviation angle is 0° . During the motion of a mechanism, these angles will of course change in value. A transmission angle of 0° occurs at a change-point position, at which point the output link, being in line

with the coupler, will not move regardless of how large a force is applied to the input link. In fact, due to friction in the pin joints, the general rule of thumb is to reject mechanisms with transmission angles of less than 30° . This limiting value will, of course, depend somewhat on the specific application for the linkage.

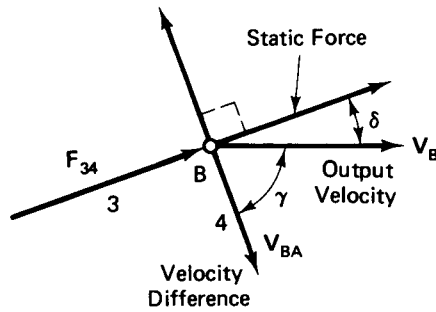


Figure 2.17 Transmission and deviation angles, γ and δ , of a four-bar linkage.

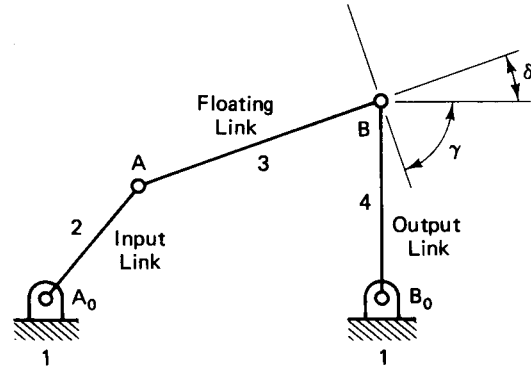


Figure 2.18 Transmission (γ) and deviation (δ) angles.

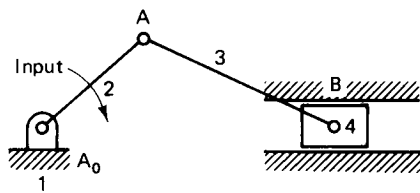


Figure 2.19 Slider-crank mechanism.

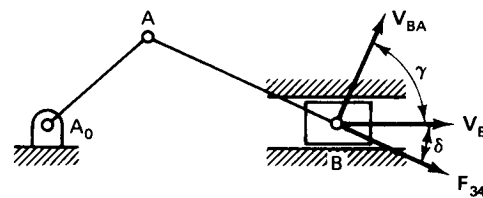


Figure 2.20 Transmission and deviation angles of the slider-crank mechanism of Fig. 2.18.

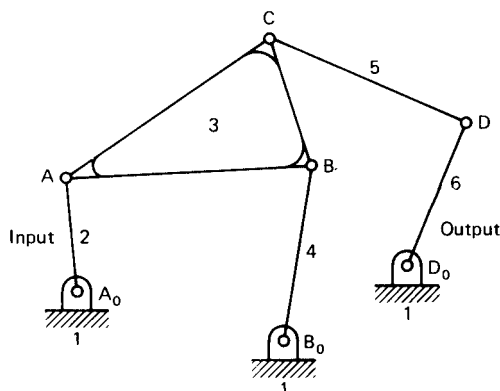


Figure 2.21 Stevenson six-bar mechanisms.

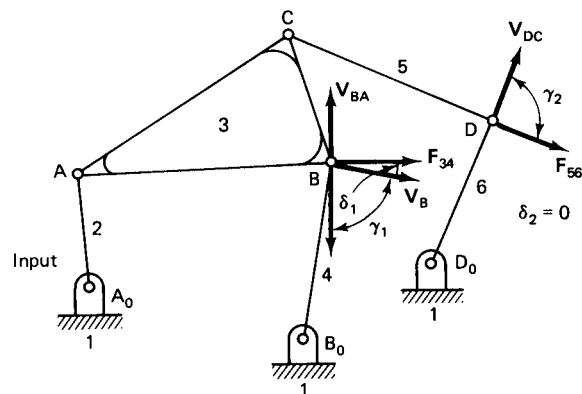


Figure 2.22 Transmission and deviation angles for six-bar with input on link 2. Note that the deviation angle at D is zero in this position.

Example 2.1

Find the transmission and deviation angles for the mechanisms in Fig. 2.19 and 2.21.

Solution In the slider-crank mechanism, the velocity of the output is along the slide and the static force $\mathbf{F}_{34}$ is along link 3, which is a two-force member. Figure 2.19 shows the resulting transmission and deviation angles for the slider-crank linkage.

The six-bar linkage (Fig. 2.21) with input on link 2 and output on link 6 will create two locations of concern for transmission and deviation angles. Four-bar A_0ABB_0 will bind up if ABB_0 lie in one line forming a dead center position, regardless of how good or bad the situation is at point D (provided there are no forces acting in the dyad of links 5 and 6). The same statement is true at point D ; There could be a 90° transmission angle at B , but if CDD_0 are lined up, the mechanism will not move. Figure 2.22 shows the set of transmission and deviation angles for this case.

It should be pointed out that if we reverse the input and output links for the six-bar in Fig. 2.20, then the analysis becomes more difficult because link 3 is no longer a two-force member. A method has been suggested [167] that creates a virtual equivalent linkage for this mechanism. The four-bar A_0ACBB_0 is replaced by a virtual link C_0C that is kinematically equivalent for position and velocity. In the next section, we will discuss instant centers and find that the extended plane of link 3 has a unique point C_0 (the instant center between links 1 and 3) that has momentarily zero velocity with respect to ground. C_0 is found (as we will see) at the intersection of the extensions of links 2 and 4. The instantaneous velocity of point C is in the same direction (and has the same magnitude) for the original six-bar as it would be for the virtual equivalent four-bar linkage D_0DCC_0 . Therefore, the transmission and deviation angles shown in Fig. 2.23 are derived. This case exemplifies the complexity of finding the transmission and deviation angles for multi-loop mechanisms. You may want to reread this section after you have studied instant centers. Reference 167 suggests a method for dealing with multi-loop mechanisms. The key is to look for locations in the mechanism where there is a possible dead center position. In addition, look for possible virtual equivalent linkages to reduce the complexity.

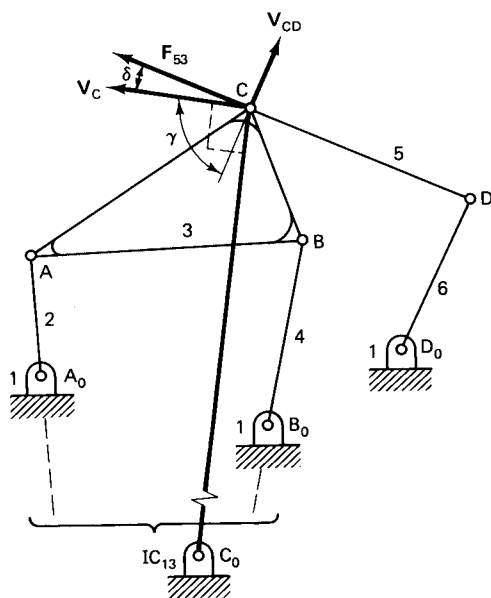


Figure 2.23 Transmission and deviation angles of the six-bar mechanism of Fig. 2.21 with input and output link reversed.

Matrix-based definitions have also been developed which measure the ability of a mechanism to transmit motion. The value of a determinant (which contains derivatives of motion variables with respect to an input motion variable for a given mechanism geometry and is called the *Jacobian*) is a measure of the ease of mobility of the mechanism in a particular position.

2.2 INSTANT CENTERS

Instant center:

A point at which there is no relative velocity between two links of a mechanism at that instant in time.

The instantaneous-center or instant-center method is a useful technique to determine relative velocity of links in a mechanism. In this section, we discuss a method for determining the instant centers of a mechanism. Later we will use instant centers to determine the mechanical advantage.

An *instantaneous center* or simply *instant center* is a point at which there is no relative velocity between two links of a mechanism at that instant. The two-link system of Fig. 2.24 consists of a link 2 and ground (link 1) which are connected together at A_0 by a pin (or revolute) joint. The point at which links 1 and 2 have no relative velocity is obviously point A_0 . In fact, for all positions in the motion of link 2, the instant center (1,2) is located at A_0 .

Notice that with the absolute velocity of one point, say A , on link 2 known, with the aid of the instant center of link 2 with respect to ground, a simple construction yields the absolute velocity of any other point, such as B . A *gauge line* is drawn from A_0 through the tip of the velocity vector V_A . There is a linear relationship between the magnitude of the velocity and the distance from the instant center (1,2) (since $V = r\omega$, where V = velocity, r = radius, and ω = angular velocity). A circular arc centered at A_0 through B locates B' on line A_0A . V'_B is drawn parallel to V'_A extending to the gauge line. Since B and B' are equidistant from A_0 , V'_B has the same magnitude as the velocity of B . Furthermore, V_B is perpendicular to the position vector A_0B and thus the velocity of B is determined.

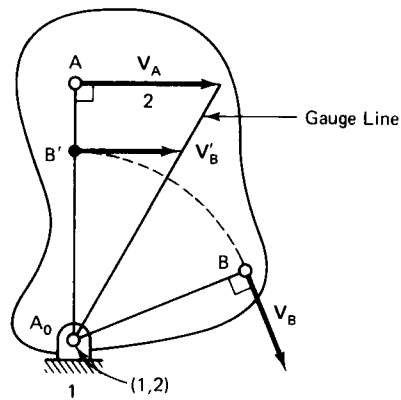


Figure 2.24 Pivot A_0 is the *instant center* for all positions of moving link 2 with respect to link 1. Graphical construction of the velocity of point B of link 2 with the aid of the instant center.

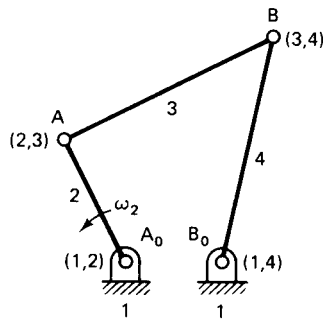


Figure 2.25 Obvious instant centers in a linked mechanism are at the pin joints between links.

In a classic four-bar linkage Fig. 2.25, we can identify several instant centers: Instant center (1,2) is located at A_0 and center (1,4) is located at B_0 . Also, instant centers (2,3) and (3,4) are located at A and B , respectively. Understand that as the linkage moves, these last two pin joints remain instant centers, but their positions do not remain fixed with respect to ground.

Are there instant centers between links 1 and 3 and links 2 and 4? Let us examine this question by focusing on the coupler link (AB) of the classic linkage of Fig. 2.26. This figure shows two finitely separated positions of the coupler link. The two positions of pins A and B are represented as A_1, A_2 , and B_1, B_2 . If the perpendicular bisector of $B_1 B_2$ is drawn, any point along this line could serve as a ground pivot which, if connected to B , would serve as the center of rotation of B from B_1 to B_2 . If the perpendicular bisector of $A_1 A_2$ is also drawn, one will observe the intersection of these bisectors at P_{12} (*pole 1,2*). This is the point which, if connected rigidly to the coupler link AB , can serve as the fixed pivot about which link AB can be rotated from position 1 to 2. The motion of the coupler link between positions would not (in general) be duplicated, but the two end positions would be exact. Now, if these two positions are brought closer and closer together, line $A_1 A_2$ will approach the direction of V_A , $B_1 B_2$ will approach that of V_B , and P_{12} will approach the instant center between links 1 and 3, (1,3). Also, the perpendicular bisectors would approach links 2 and 4, respectively. Thus the instant center $I_{1,3}$ will be at the intersection of the extension of links 2 and 4. This construction is valid when the two links of interest (1 and 3) are connected by binary links (2 and 4). But how can we find the location of less obvious instant centers?

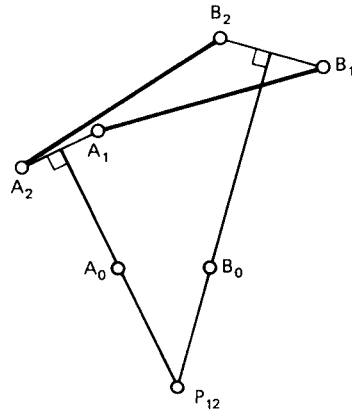


Figure 2.26 The pole 12 (P_{12}) about which the coupler AB could rotate from position 1 to 2.

Kennedy's Theorem

Kennedy's Theorem: The three instantaneous centers of three bodies moving relative to one another must lie along a straight line.

Kennedy's theorem deals with the three instant centers between three links of a system of rigid members. In Fig. 2.27 there are three links: ground (link 1) and two other links (2 and 3) which are pinned to ground. Link 3 passes behind link 2. Link 2 and link 3 have angular velocities ω_2 and ω_3 relative to link 1, as shown. The instant centers (1,2) and (1,3) are located at the respective pin joints, but where is instant center (2,3)? For instance, is point P the instant center? Think of P first as being part of link 2, (P_2) and then as part of link 3, (P_3). P_2 has an absolute velocity which is perpendicular to O_2P_2 while P_3 has an absolute velocity which is perpendicular to O_3P_3 . Now, if P were the instant center I_{23} , then $V_{P_2P_3} = -V_{P_3P_2}$ would be zero. Notice that point P could not be the instant center (2,3) because the directions of $V_{(P_2)}$ and $V_{(P_3)}$ are not the same. Where, at least, would the *directions* of the two absolute velocity vectors be the same? At any point along the line of instant centers (1,2) and (1,3) (say at point Q) the absolute velocities of that point, either as a point on link 2 or on link 3, would be perpendicular to the line between these centers. Depending on the angular velocities ω_2 and ω_3 , the magnitudes will be equal somewhere along the line of centers or its extensions. This brings us to Kennedy's theorem of three centers: The three instantaneous centers of three bodies moving relative to one another must lie along a straight line (see Fig. 2.28).

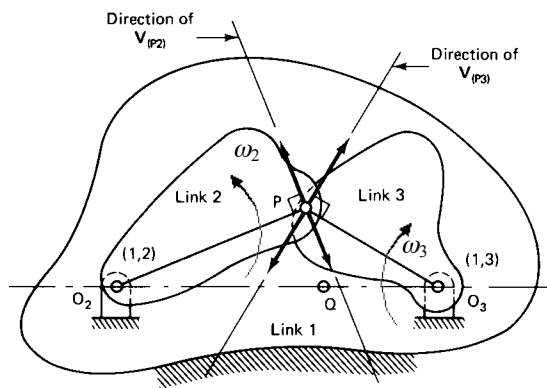


Figure 2.27 Three links (1, 2, and 3) moving with respect to each other. Where is the instant center of links 2 and 3?

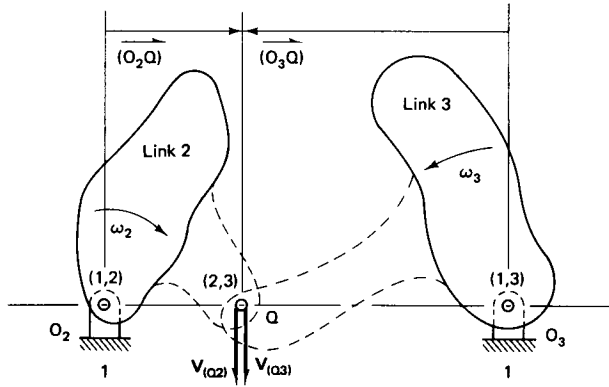


Figure 2.28 Proof of Kennedy's theorem that all three instant centers of three links moving with respect to one another must lie on a straight line.

Before returning to the classic four-bar linkage in Fig. 2.25 with this theorem, let us develop Kennedy's theory further. Suppose that the instant center (2,3) were located at point Q, as shown in Fig. 2.28; then

$$\mathbf{V}_{(Q2)} = i\omega_2 (\overrightarrow{O_2Q}) = i\omega_2 (\overrightarrow{1,2 - 2,3}) \quad (2.1)$$

$$\mathbf{V}_{(Q3)} = i\omega_3 (\overrightarrow{O_3Q}) = i\omega_3 (\overrightarrow{1,3 - 2,3}) \quad (2.2)$$

but

$$\mathbf{V}_{(Q2)} = \mathbf{V}_{(Q3)} \quad (2.3)$$

so that angular velocity ratio is

$$\frac{\omega_2}{\omega_3} = \frac{(\overrightarrow{O_3Q})}{(\overrightarrow{O_2Q})} = \frac{(\overrightarrow{1,3 - 2,3})}{(\overrightarrow{1,2 - 2,3})} \quad (2.4)$$

which is a positive or negative real, which depends on the direction of the vector on the right side of this equation. The absolute value of Eq. (2.4) is

$$\left| \frac{\omega_2}{\omega_3} \right| = \frac{(\overline{1,3 - 2,3})}{(\overline{1,2 - 2,3})} \quad (2.5a)$$

or in general

$$\left| \frac{\omega_x}{\omega_y} \right| = \frac{(\overline{1,y - x,y})}{(\overline{1,x - x,y})} \quad (2.5b)$$

If the relative instant center (2,3) lies between the absolute instant centers (1,2 and 1,3) the angular velocity ratio is negative (i.e., the two links rotate in opposite directions). If the relative instant center lies outside the other two, the angular velocity ratio is positive (i.e., links rotate in the same direction). Equation (2.5) relates an angular velocity ratio to a ratio of distances between instant centers.

Another way of looking at instant centers involves replacing a linkage instantaneously with a kinematically equivalent pair of gears. Figure 2.29 shows a gear pair that produces instantaneously, the same angular velocity ratio as that of links 2 and 3 in Fig. 2.28. The pitch radius vector of gear 2 is $(1,2 - 2,3)$, and $(1,3 - 2,3)$ is the pitch radius vector of gear 3. A positive angular velocity ratio case is

depicted in Fig. 2.30. In summary, if the *relative instant center* (2,3) [This is the instant center that does not include ground – the number “1”] is between the *absolute instant centers* (1,2) and (1,3), then the angular velocity ratio is negative. If the relative instant center is not between the absolute instant centers, then the angular velocity ratio is positive.

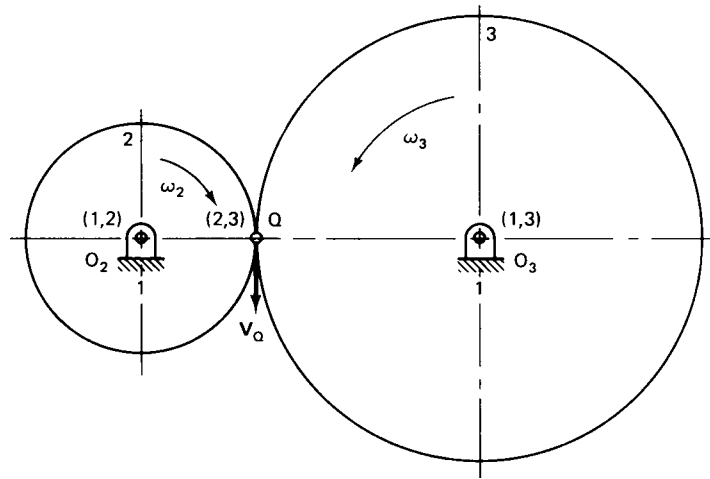


Figure 2.29 Instant centers of a gear pair with negative velocity ratio. This gear pair would generate the same angular velocity ratio (ω_2/ω_3) as links 2 and 3 in Fig. 2.28 if the instant centers were in identical positions.

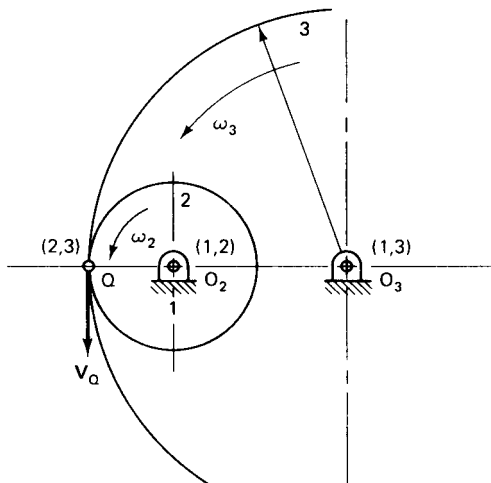


Figure 2.30 Instant centers of a gear pair with positive velocity ratio.

Returning to the classic four-bar linkage of Fig. 2.25, we now are able to find the locations of instant centers (2,4) and (1,3). According to Kennedy's theorem, instant center (1,3) lies on a line drawn through the common instant centers between links 1, 3, and another link. If that other link is link 2, instant centers (2,3), (1,2), and (1,3) must lie on a straight line; so a line is drawn between centers (2,3) and (1,2) (see Fig. 2.31). If links 1, 3, and 4 are utilized, centers

(3,4), (1,4), and (1,3) must lie on a straight line. A line through (3,4), (1,4) intersects the other line, locating center (1,3). An Instant center (2,4) is located by drawing lines through centers (2,3), (3,4) and through centers (1,2), (1,4). Compare this procedure with the intuitive method described by going to the limit (infinitesimal displacement) of the coupler link in Fig. 2.26. Observe that pole P_{12} in Fig. 2.26 approximates the location of the instant center (1,3) in Fig. 2.31.

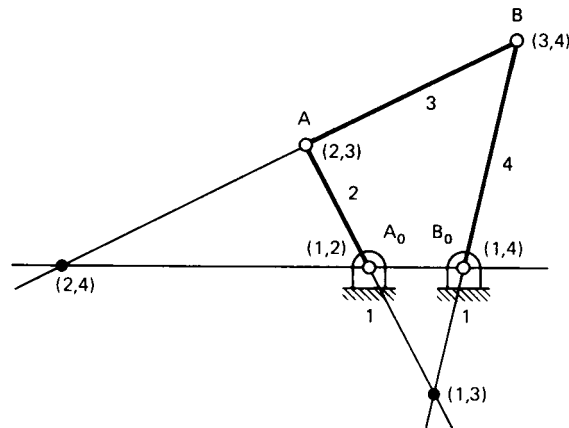


Figure 2.31 Construction of instant centers between opposite links of a four-bar mechanism.

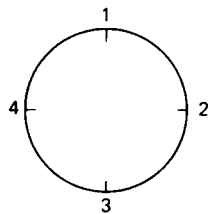


Figure 2.32 Successive steps in keeping track of instant center constructions in Fig. 2.31.

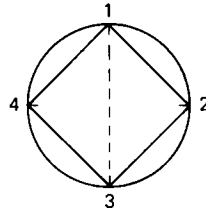


Figure 2.33 Successive steps in keeping track of instant center constructions in Fig. 2.31.

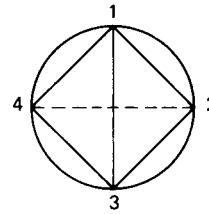


Figure 2.34 Successive steps in keeping track of instant center constructions in Fig. 2.31.

The instant center between two links that are not connected directly together can be thought of as an instantaneous pin joint between those links. For example, we could pin link 3 to ground (link 1) at instant center (1,3) and obtain the same instantaneous relative velocity between those links as the four-bar mechanism has provided. The concept of lower-pair equivalency introduced in Table 1.2 is based on this velocity-equivalence concept.

There is a short graphical method (based on the principle of duality in graph theory) that helps keep track of the instant centers already obtained and those which are obtainable. Although this method may not be warranted for a simple linkage such as a four-bar, it is extremely useful in more complex mechanisms. A circle is drawn and divided by a number of tic marks equal to the number of links in the mechanism. For the classic four-bar linkage, four tic marks are labeled to

represent the four links (see Fig. 2.32). Once an instant center is obtained, a line is drawn between the two numbered tics to account for that located instant center. The tics represent the links, and the line represents a known instant center between two links.

Thus one line represents one instant center. For example, once centers (1,2), (2,3), (3,4) and (1,4) are obtained in Fig. 2.31, solid lines connect 1, 2; 2, 3; 3, 4; and 1, 4 in Fig. 2.33. To obtain additional centers using Kennedy's theorem, we need to find intersections of lines containing appropriate instant centers in Fig. 2.31. The graphical accounting procedure of Figs. 2.32, 2.33, and 2.34 reminds us not only of which centers have not yet been obtained, but which are immediately available and which intersections are appropriate. To find an instant center (x,y) between links x and y, draw a dashed line connecting x and y. So, if we wish to find (1,3), a dashed line is drawn between 1 and 3 (Fig. 2.33). Since there are two sets of two solid lines connecting 1 and 3, center (1,3) is attainable. The diagram (and Kennedy's theorem) says that a line through (1,4) and (3,4) together with a line through (1,2) and (2,3) will intersect at (1,3). Once this is completed, as in Fig. 2.30, a solid line replaces the dashed line between 1 and 3 (see Fig. 2.34). The dashed line in Fig. 2.34 indicates that center (2,4) is attainable by lines through (1,4), (1,2) and (3,4), (2,3), as demonstrated in Fig. 2.31.

Before going to a more complex mechanism, it is worthwhile to note that the number of instant centers N of a mechanism with n links is

$$N = \frac{n(n-1)}{2} \quad (2.6)$$

This is intuitively true, because each of the n links has an instant center common with each one of the other $(n-1)$ links, but center (j,k) is the same as (k,j) ; hence the product $n(n-1)$ must be divided by 2.

Example 2.2

Determine the locations of all the instant centers for a six-bar mechanism in Fig. 2.34 in preparation for a velocity analysis of this mechanism.

Solution The number of instant centers is, by Eq. (2.6), is 15.

The instant centers, which are obtained by inspection, are shown in the circular diagram (Fig. 2.35). Notice that the instant center between the slider (link 6) and the slide (link 5) is off at infinity in the direction perpendicular to the slide. The circular diagram (Fig. 2.35) indicates that several instant centers are now available (this is not the general rule with more complex mechanisms; usually one at a time is attainable, thus enabling the determination of the next). Centers (1,3), (1,5), (2,4), and (4,6) can be determined as described above. Notice particularly the construction of center (4,6). Figure 2.36 dictates that (4,6) lies along lines connecting (1,6), (1,4) and (5,6), (4,5). In the second pair, (5,6) is at ∞ in the direction perpendicular to link 5. How can a line be drawn through (5,6) and (4,5)? Simply draw a line parallel to (1,6)–(5,6) through (4,5), as shown in Fig. 2.37. This can be justified if you picture yourself sitting at (4,5) and looking at (5,6) off in infinity in the direction of (1,6)–(5,6). It is clear that your line of sight would be parallel to (1,6)–(5,6).

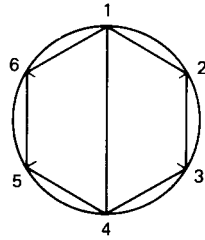


Figure 2.35 Obvious instant centers in a six-bar mechanism.

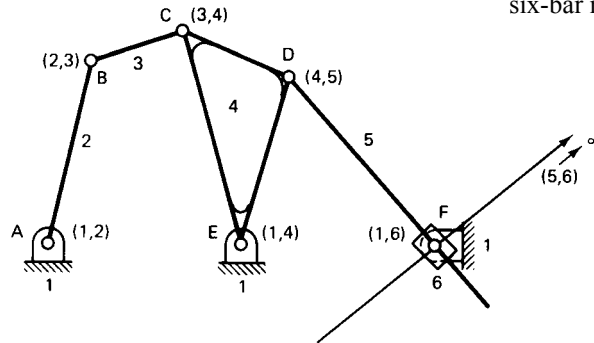


Figure 2.38 indicates the stage of the analysis where (4,6), (1,5), (2,4), and (1,3) have been obtained. Instant center (2,5) is one possible next choice. The intersection of lines through (1,2), (1,5) and (2,4) (4,5) yields this center, which will be used in Example 2.4 8for velocity analysis.

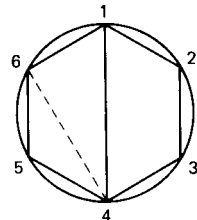


Figure 2.36 The dashed line between 4 and 6 indicates the desire to find instant center (4,6). Two sets of lines $[(1,6) - (1,4)]$ and $[(4,5) - (5,6)]$ tell us that (4,6) is obtainable.

The rest of the instant centers — (3,5), (2,6), (3,6) — are obtained following the same procedure (see Fig. 2.37). Notice that these last three centers have more than two intersecting lines which locate them. This provides a redundant verification indicating the accuracy of the construction. If more than two lines intersect at the same point, everything is fine. A slight discrepancy indicates a slight inaccuracy in the drawing. If the intersections are not close, a major mistake has been made and a complete review of previous steps is warranted.

Mechanical Advantage:

The ratio of input/output torque changes during the motion cycle a linkage.

$$MA = \frac{F_{out}}{F_{in}}$$

Torque Ratio

is a function of angular velocity Ratio and Geometric Parameters.

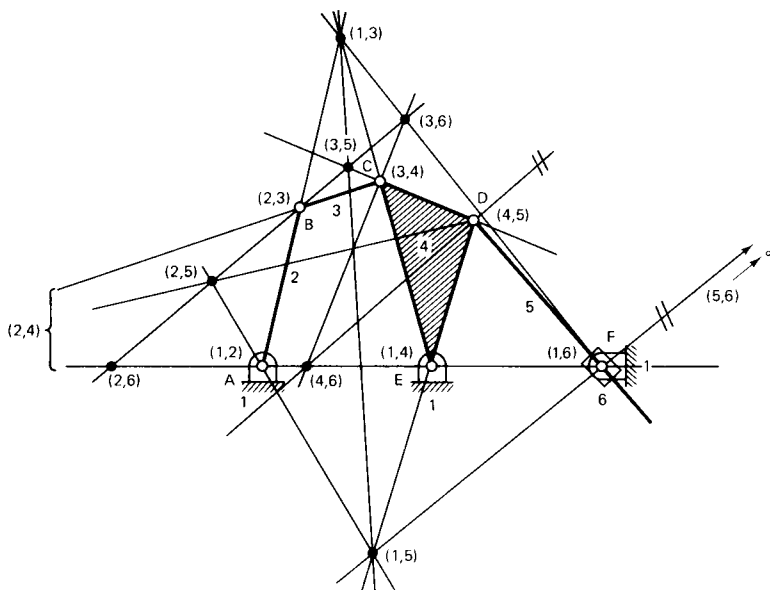


Figure 2.37 Construction of all instant centers for the six-link mechanism of Fig. 2.35. Figures 2.36 and 2.38 show several stages of the graphical “bookkeeping” during the construction.

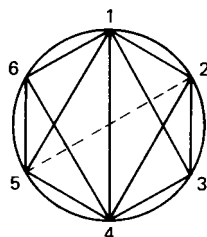


Figure 2.38 Instant center (2,5) can be found by intersections of lines $(1,2) - (1,5)$ and $(2,4) - (4,5)$.

2.3 MECHANICAL ADVANTAGE

The ability of a mechanism to transmit torque or force from input to the output is an important criterion in design. Some mechanisms, such as a gear train, transmit a constant torque ratio between the input and output because there is a constant speed ratio between input and output. In a linkage, however, this ratio changes during the motion cycle in a linkage. How might we determine a relationship between the input and output forces? Two observations can be made without further analysis.

1. The torque ratio is a function of the speed or angular velocity ratio between output and input links of the mechanism.
2. The torque ratio is a function of geometric parameters, which, in the case of a linkage, will generally change during the course of the mechanism motion.

If we assume that a mechanism is a conservative system (i.e., energy losses due to friction, heat, etc., are negligible compared to the total energy transmitted by the system), and if we assume that there are no effects of inertia forces, then

Conservative Mechanical System:

Energy losses due to friction, heat, wear, etc. are negligible compared to the energy transmitted.

the input power P_{in} is equal to the output power P_{out} . Thus the input torque times the angular velocity of the input is equal to the output torque times the angular velocity of the output:

$$P_{in} = T_{in} \omega_{in} = T_{out} \omega_{out} = P_{out} \quad (2.7)$$

or,

$$P_{in} = \mathbf{F}_{in} \cdot \mathbf{V}_{in} = \mathbf{F}_{out} \cdot \mathbf{V}_{out} = P_{out} \quad (2.8)$$

where T_{in} and F_{in} are torque and force exerted *on* the linkage, and T_{out} and F_{out} are those exerted *by* the linkage; and where $\mathbf{V}_{in}$ and $\mathbf{V}_{out}$ are the velocities of the points through which $\mathbf{F}_{in}$ and $\mathbf{F}_{out}$, respectively, act; and where, in vector form,

$$\mathbf{V} \cdot \mathbf{F} = VF \cos(\arg \mathbf{F} - \arg \mathbf{V}) \quad (2.9a)$$

Also,

$$\mathbf{V} \cdot \mathbf{F} = V_x F_x + V_y F_y \quad (2.9b)$$

Figure 2.39 reminds us that neither force (torque) nor velocity alone is constant through a linkage mechanism. The misguided designers of so-called *perpetual motion machines* disregard this fact.

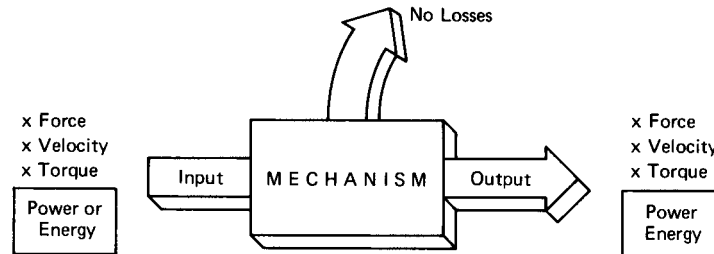


Figure 2.39 Power and energy are conserved through the mechanism. Force, velocity, and torque are not.

Notice that the units of torque times angular velocity and the scalar product of force and velocity both represent power. From Eq. (2.7),

$$\frac{T_{out}}{T_{in}} = \frac{\omega_{in}}{\omega_{out}} \quad (2.10)$$

By definition, *mechanical advantage* (M.A.) is the ratio of the magnitudes of force-out over force-in:

$$\text{M.A.} = \frac{F_{out}}{F_{in}} \quad (2.11)$$

where $F = |\mathbf{F}|$.

Combining Eqs. (2.11) and (2.7) by noticing that torque is the product of force times a radius,

$$\text{M.A.} = \left(\frac{T_{\text{out}}}{r_{\text{out}}} \right) \left(\frac{r_{\text{in}}}{T_{\text{in}}} \right) = \left(\frac{r_{\text{in}}}{r_{\text{out}}} \right) \left(\frac{T_{\text{out}}}{T_{\text{in}}} \right) \quad (2.12)$$

and

$$\boxed{\text{M.A.} = \left(\frac{r_{\text{in}}}{r_{\text{out}}} \right) \left(\frac{\omega_{\text{in}}}{\omega_{\text{out}}} \right)} \quad (2.13)$$

Mechanical advantage **depends only on the geometry of the linkage.** It does not change when the mechanism is actuated at different rates.

Thus the mechanical advantage is a product of two factors: (1) a ratio of distances that depend on the placement of the input and output forces and (2) an angular velocity ratio. The first factor may not change in value as the mechanism moves, but the second one will change in most linkage mechanisms. Since the angular velocity ratio can be expressed entirely in terms of directed distances (based on the instant-center development), the mechanical advantage can be expressed entirely in terms of ratios of distances.

Let us look at the four-bar mechanism in Fig. 2.40. If we neglect the weight of links 2, 3, and 4, what reading would you expect the scale to display as a result of the block weighing 10 Nm on link 2 of the mechanism? Using Eq. (2.13),

$$\frac{F_{\text{out}}}{F_{\text{in}}} = \text{M.A.} = \left(\frac{\omega_{\text{in}}}{\omega_{\text{out}}} \right) \left(\frac{r_{\text{in}}}{r_{\text{out}}} \right) \quad (2.14)$$

In this case link 2 is the input while link 4 is the output. Thus from equation 2.5b,

$$\text{M.A.} = \frac{F_{\text{out}}}{F_{\text{in}}} = \left(\frac{\omega_2}{\omega_4} \right) \left(\frac{r_{\text{in}}}{r_{\text{out}}} \right) = \frac{(1,4-2,4)}{(1,2-2,4)} \frac{(r_{\text{in}})}{(r_{\text{out}})} \quad (2.14a)$$

Note that the common instant center (2,4) is outside the others, making the angular velocity ratio positive. Measuring distances on Fig. 2.40 and solving for F_{out} ,

$$F_{\text{out}} = F_{\text{in}} (\text{M.A.}) = (10) \frac{(2)}{(.5)} \frac{(1.5)}{(.5)} = (10)(4)(3) = 120 \text{ lbf}$$

thus the M.A. is $(4)(3) = 12$. The gain in mechanical advantage has contributions both from the radius ratio and the angular velocity ratio. Both are distances measured on the diagram.

This result can be verified by way of free-body diagrams of Fig. 2.41. Here too, the mechanical advantage is purely in terms of distances.

$$\text{M.A.} = \left(\frac{1.5}{0.3} \right) \left(\frac{1.2}{0.5} \right) = (5)(2.4) = 12$$

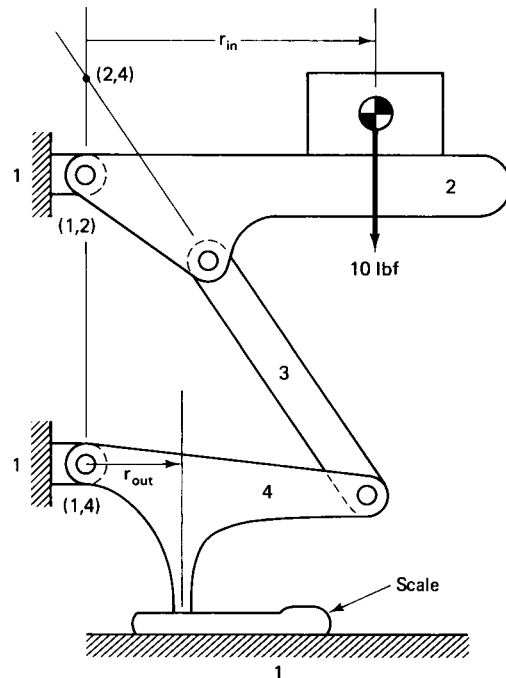


Figure 2.40 Determination of the scale reading based on a weight of 10 lbf.

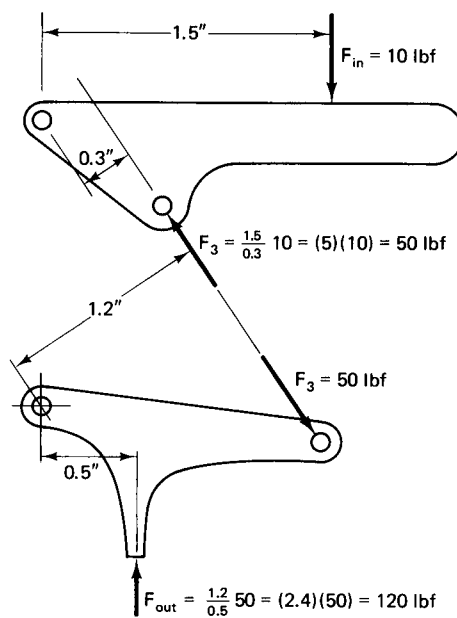


Figure 2.41 Determination of mechanical advantage by way of free-body diagrams.

Expressions in the form of Eq. (2.14) are *powerful design tools* and can usually be verified by inspection. In many design situations, the mechanical advantage expression for a mechanism will allow the optimal redesign of that device for improved mechanical advantage. Practical considerations such as the maximum permitted size of the mechanism will usually limit the amount of change allowable from an original design. The case study at the end of Chapter 8 illustrates this point.

For example, suppose that the four-bar linkage of Fig. 2.42 is being used as the driving mechanism of a manually operated pump. In the position shown, the

handle is being pulled left with force F_{in} . Meanwhile, the pressure difference across the piston in the cylinder is resisting the movement by a force equal and opposite to F_{out} . What is the mechanical advantage of this device in the position shown? (The piston rod is instantaneously perpendicular to link 4.)

If the input link is identified as link 2 and the output link as link 4, then according to Eq. (2.10),

$$\frac{T_4}{T_2} = \frac{\omega_2}{\omega_4}$$

Instant center (2,4) is found to lie between (1,2) and (1,4), so that

$$\frac{T_4}{T_2} = \frac{\omega_2}{\omega_4} = \frac{\overrightarrow{(1,4-2,4)}}{\overrightarrow{(1,2-2,4)}} \quad (2.15)$$

$$\text{M.A.} = \left(\frac{r_{in}}{r_{out}} \right) \frac{\overrightarrow{(1,4-2,4)}}{\overrightarrow{(1,2-2,4)}} \quad (2.16)$$

Notice the factors that make up the mechanical advantage. The mechanical advantage is greater if the ratio z_{in}/z_{out} is greater. This checks intuitively with Fig. 2.42. The second ratio can be checked intuitively also. Notice that as point P is moved to the left (link 2 rotated counterclockwise) to the position shown in Fig. 2.43, where the four-bar linkage is in its “toggle” position, centers (1,2) and (2,4) coincide. According to Eq. (2.16), the mechanical advantage goes to infinity in this position. Since links 2 and 3 line up, (ideally) no force is required at P to resist an infinite force at Q . Of course, bending of link 4 would occur before an infinite force could be applied.

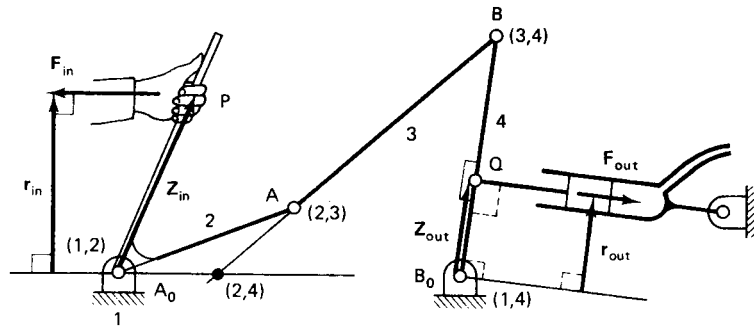


Figure 2.42 Mechanical advantage of this four-bar hand pump drive mechanism increases as the toggle position (A_0 , A , and B collinear) is approached.

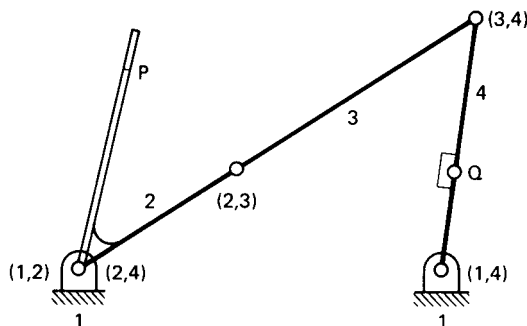


Figure 2.43 Toggle position of the pump drive mechanism of Fig. 2.41: M.A. is theoretically infinite: no force at P can overcome a very large force at Q .

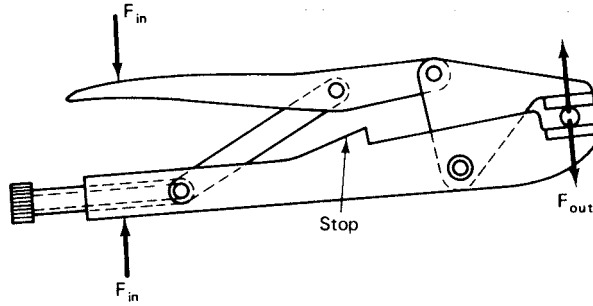


Figure 2.44 The stop prevents excessive overtravel beyond the toggle position of the upper handle and the coupler link.

Mechanical advantage of this device can be increased by one or more of the following:

1. Increase z_{in} .
2. Decrease z_{out} .
3. Move B_0 away from (2,4) (keeping the rest of the linkage the same).
4. Move A_0 toward (2,4) (keeping the rest of the linkage the same).
5. Move point A until links 2 and 3 line up.
6. Make $\mathbf{F}_{in}$ perpendicular to $\mathbf{z}_{in}$.

Example 2.3

Determine the mechanical advantage of the adjustable toggle pliers in Fig. 2.43. Why is this device designed this way?

Solution First, the instant centers of the pliers are found. Let us designate input link = 3, output link = 4, and ground link = 1 (see Fig. 2.45). Therefore instant centers (1,3), (3,4), and (1,4) are of most interest. Also $T_3\omega_3 = T_4\omega_4$, since $\mathbf{F}_{out}$ is perpendicular to $\mathbf{r}_{out}$ and $\mathbf{F}_{in}$ is perpendicular to $\mathbf{r}_{in}$:

$$\begin{aligned} T_4 &= [\mathbf{r}_{out} \times \mathbf{F}_{out}] \\ &= r_{out} F_{out} \sin(\arg \mathbf{F}_{out} - \arg \mathbf{r}_{out}) = r_{out} F_{out} \sin(-90^\circ) = -1.9 F_{out} \end{aligned}$$

Similarly,

$$T_3 = [\mathbf{r}_{in} \times \mathbf{F}_{in}] = (5.1) F_{in}$$

From Eq. (2.10),

$$\frac{T_4}{T_3} = \frac{\omega_3}{\omega_4} = \frac{\overrightarrow{(3,4-1,4)}}{\overrightarrow{(3,4-1,3)}} = \frac{-1.6}{0.7} = -2.29$$

from which

$$\text{M.A.} = \frac{F_{out}}{F_{in}} = \left| \frac{r_{in}}{r_{out}} \frac{\omega_3}{\omega_4} \right| = \left| \frac{5.1}{1.9} (-2.29) \right| = 6.15$$

In the position shown, the pliers have only a 6.15:1 mechanical advantage. As we clamp down on the pliers, however, instant center (2,4) approaches center (1,2). When (1,2), (2,3), and (3,4) are close to being in a straight line, (2,4) is nearly coincident with (1,2), (1,3) approaches (3,4), and the mechanical advantage approaches infinity. The plier position of Figure 2.46

is graphed in Figure 2.45. It depicts the change in mechanical advantage as link 3 is rotated counterclockwise. Continued rotations will create infinite MA as instant center (1,3) locates itself on top of (3,4).

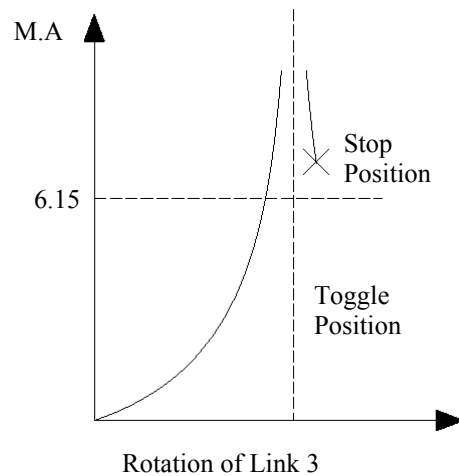


Figure 2.45 M.A. vs. Rotation of link 3

The screw adjustment should be set so that the maximum mechanical advantage occurs at the required distance between the jaws of the pliers. In fact, in some brands, there is a stop located in the “over center” position (just past the dead center), as shown in Figs. 2.44 and 2.45. This gives both a very high mechanical advantage and stable “grip” for the linkage, since it would take an ideally infinite force at the jaws to move the linkage back through its toggle position.

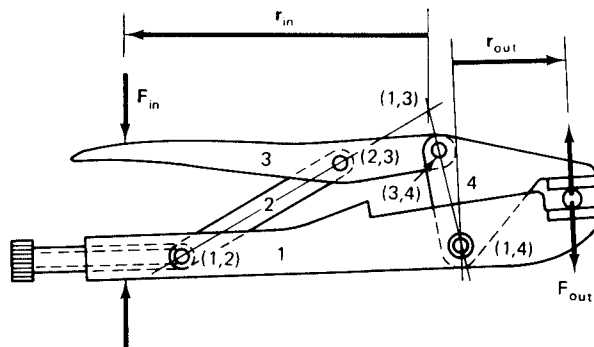


Figure 2.46 As (3,4), (2,3), and (1,2) approach collinearity, M.A. approaches infinity.

2.4 ANALYTICAL METHOD FOR VELOCITY AND MECHANICAL ADVANTAGE DETERMINATION

The procedures we’ve described so far, for velocity and mechanical advantage analysis are basically graphical solution procedures. Some analytical, complex-number equivalents have been presented. If only a finite number of positions of a linkage are to be analyzed, any of the graphical procedures are certainly useful. If, however, a large number of positions and/or linkages need to be analyzed, the graphical procedures are too time consuming. With desktop or laptop computers, programmable calculators, or software such as Matlab readily available, the analytical expressions based on complex-number representation are extremely

valuable. After one becomes comfortable with the graphical procedures (gaining a clearer conceptual image of vector solutions), purely analytical methods can be used with greater confidence. When a question arises from the non-visual techniques, a graphical spot check can verify the results. Also, computer graphics can be used to visually display the results of analytical methods (see Sec. 3.11 and the color inserts in this book).

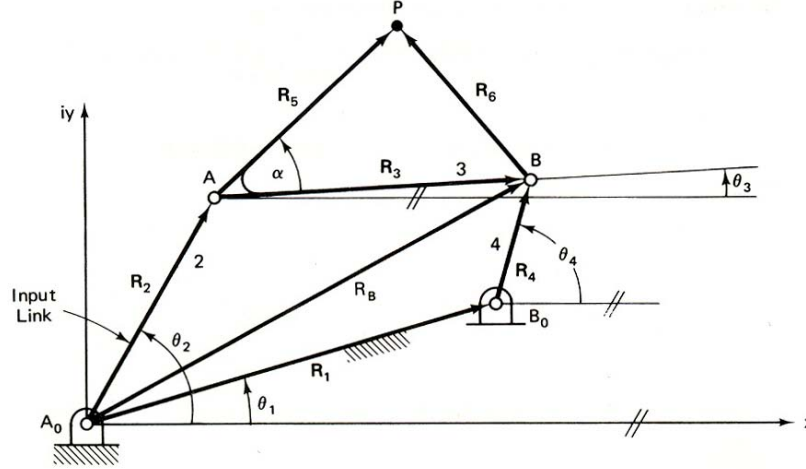


Figure 2.47 Vector notation for velocity analysis by complex numbers in four-bar mechanism.

Figure 2.47 shows a vector representation of a four-bar linkage, the angular velocities of links 3 and 4 may be determined as functions of the input link (link 2) angular velocity and the displacement position parameters. Recall that a displacement analysis for this same linkage was presented in Sec. 2.2.

The position vector from the base of the input link (A_0) to point B may be written via two routes:

$$\mathbf{R}_B = \mathbf{R}_2 + \mathbf{R}_3 \quad (2.17)$$

$$\mathbf{R}_B = \mathbf{R}_1 + \mathbf{R}_4 \quad (2.18)$$

The derivatives of these expressions will yield the velocity of point B . Using the polar form of these expressions for taking time derivatives leads to

$$\mathbf{R}_B = r_2 e^{i\theta_2} + r_3 e^{i\theta_3} = r_1 e^{i\theta_1} + r_4 e^{i\theta_4} \quad (2.19)$$

$$\mathbf{V}_B = r_2 \omega_2 i e^{i\theta_2} + r_3 \omega_3 i e^{i\theta_3} = r_4 \omega_4 i e^{i\theta_4} \quad (2.20)$$

Now using the Cartesian form of Eq. (2.20) to break it into real and imaginary parts, we obtain

$$r_2 \omega_2 \sin \theta_2 + r_3 \omega_3 \sin \theta_3 = r_4 \omega_4 \sin \theta_4 \quad (2.21)$$

$$r_2 \omega_2 \cos \theta_2 + r_3 \omega_3 \cos \theta_3 = r_4 \omega_4 \cos \theta_4$$

These two scalar equations contain only two scalar unknowns, ω_3 and ω_4 . Multiplying the first equation by $\cos \theta_4$ and the second by $\sin \theta_4$, we have

$$\begin{aligned} r_2 \omega_2 \sin \theta_2 \cos \theta_4 + r_3 \omega_3 \sin \theta_3 \cos \theta_4 &= r_4 \omega_4 \sin \theta_4 \cos \theta_4 \\ r_2 \omega_2 \cos \theta_2 \sin \theta_4 + r_3 \omega_3 \cos \theta_3 \sin \theta_4 &= r_4 \omega_4 \cos \theta_4 \sin \theta_4 \end{aligned} \quad (2.22)$$

Subtracting the first from the second eliminated ω_4 :

$$r_2 \omega_2 (\cos \theta_2 \sin \theta_4 - \sin \theta_2 \cos \theta_4) + r_3 \omega_3 (\cos \theta_3 \sin \theta_4 - \sin \theta_3 \cos \theta_4) = 0$$

or

$$r_2 \omega_2 \sin(\theta_4 - \theta_2) + r_3 \omega_3 \sin(\theta_4 - \theta_3) = 0 \quad (2.23)$$

Thus

$$\boxed{\omega_3 = -\frac{r_2}{r_3} \omega_2 \frac{\sin(\dot{\theta}_4 - \dot{\theta}_2)}{\sin(\dot{\theta}_4 - \dot{\theta}_3)}} \quad (2.24)$$

Eliminating the terms containing ω_3 instead of ω_4 from eq. (2.23) yields

$$\boxed{\omega_4 = \frac{r_2}{r_4} \omega_2 \frac{\sin(\dot{\theta}_3 - \dot{\theta}_2)}{\sin(\dot{\theta}_3 - \dot{\theta}_4)}} \quad (2.25)$$

These two expressions are easily programmed for automatic computation. Notice that Eqs. (2.24) and (2.25) may also be utilized for mechanical advantage analysis, where ω_3/ω_2 are utilized in the M.A. expressions.

Example 2.4

Determine the analytical expression for the velocity of point P of Fig. 2.47.

Solution The position and velocity of point P are expressed in vector form as

$$\mathbf{R}_P = r_2 e^{i\dot{\theta}_2} + r_5 e^{i(\dot{\theta}_3 + \alpha)} \quad (2.26)$$

The derivative of this expression yields

$$\mathbf{V}_P = r_2 \omega_2 i e^{i\dot{\theta}_2} + r_5 \omega_3 i e^{i(\dot{\theta}_3 + \alpha)} \quad (2.27)$$

or in Cartesian form

$$\text{Real part: } V_{Px} = -r_2 \omega_2 \sin \theta_2 - r_5 \omega_3 \sin(\theta_3 + \alpha) \quad (2.28)$$

$$\text{Imaginary part: } V_{Py} = r_2 \omega_2 \cos \theta_2 + r_5 \omega_3 \cos(\theta_3 + \alpha)$$

where ω_3 is found by Eq. (2.24).

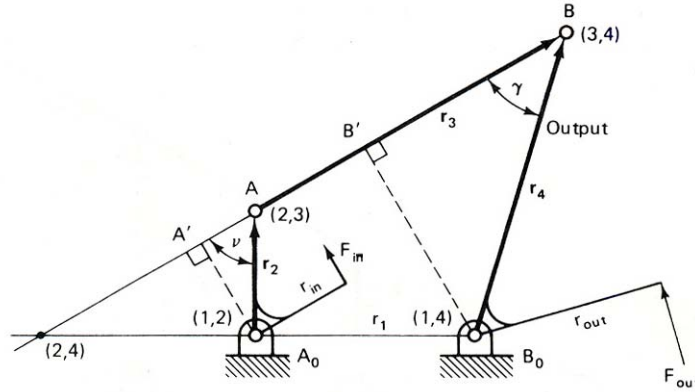


Figure 2.48 Correlation of M.A. with transmission angle γ and crank-coupler angle ν in a four-bar mechanism. See Eq. (2.33).

Correlation of Mechanical Advantage and Transmission Angle

In Sec. 2.1 we observed that the transmission angle is a means of determining the effectiveness by which motion is imparted to the output link of a particular mechanism. Further, mechanical advantage was defined as the instantaneous magnitude ratio of the output force to the input force of a particular mechanism (Sec. 2.3). Both of these indices of performance are static parameters (the effect of inertia is not included) that help us compare one linkage with another, or one position of a linkage with another position of the same linkage. Both of these indices can be expressed as a function of the geometry of the linkage. A comparison is warranted to avoid confusion of these terms.

In Fig. 2.48 the mechanical advantage may be expressed as [Eq. (2.14a)]
Constructing lines A_0A' and B_0B' perpendicular to the line containing (2,4), A , and B shows that by similar triangles,

$$\frac{(2,4-1,4)}{(1,4-B')} = \frac{(2,4-1,2)}{(1,2-A')} \quad (2.29)$$

If λ and ν are the magnitudes of the smaller of the two angles made by the coupler (or its extension) with the output and input links, respectively, then $(1,4-B') = r_4 \sin \lambda$, $(1,2-A') = r_2 \sin \nu$, and

$$\frac{(2,4-1,4)}{(2,4-1,2)} = \frac{r_4 \sin \lambda}{r_2 \sin \nu} \quad (2.30)$$

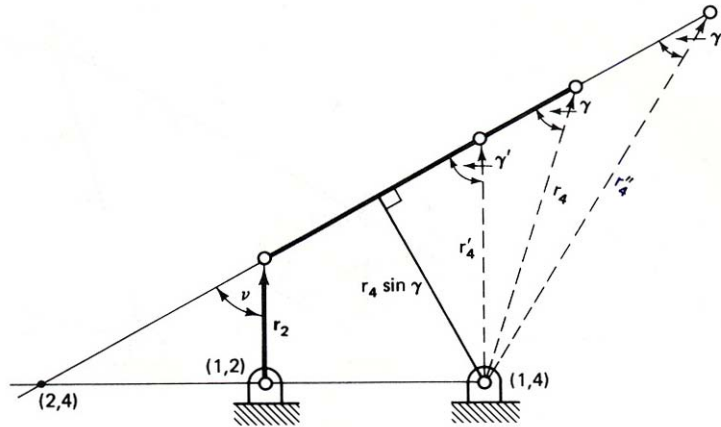


Figure 2.49 If (2,4), (1,2), (1,4) and r_2 are unchanged, variation of transmission angle does not change the M.A.

Mechanical advantage and transmission angle are related to one another. However, they give slightly different information. Which one is used may depend on the particular problem being addressed.

Thus

$$\text{M.A.} = \frac{r_{\text{in}}}{r_{\text{out}}} \left(\frac{r_4 \sin \gamma}{r_2 \sin \nu} \right) \quad (2.31)$$

Notice that the mechanical advantage becomes infinite when angle ν is either 0° or 180° . This agrees with the analysis of Sec. 2.3 since in either of these cases instant center (2,4) is coincident with center (1,2). Notice also the relationship between the transmission angle γ and the mechanical advantage (disregarding the ratio $r_{\text{in}}/r_{\text{out}}$): If angle γ is 0° or 180° , the mechanical advantage is zero. For other values of mechanical advantage, the transmission angle will vary. In Fig. 2.49, let $(2,4 - 1,4)$, $(2,4 - 1,2)$, r_2 , and ν be fixed. The transmission angle γ and the length of the output link r_4 may have different values for a given value of M.A. as long as the product $(r_4 \sin \gamma)$ remains constant. The same amount of torque will be transferred to the output link in each of the cases shown in Fig. 2.47, but for the cases with smaller transmission angles, a large component of the static force transmitted through the coupler will result in a larger bearing force at (1,4) rather than in a usable force perpendicular to the output link. Thus a linkage with good mechanical advantage may have an unacceptable transmission angle and a linkage with an excellent transmission angle in a particular position may not have a sufficient mechanical advantage. Since both the transmission angle and mechanical advantage vary with linkage position, either parameter can be critical to the designer for a given circumstance.